

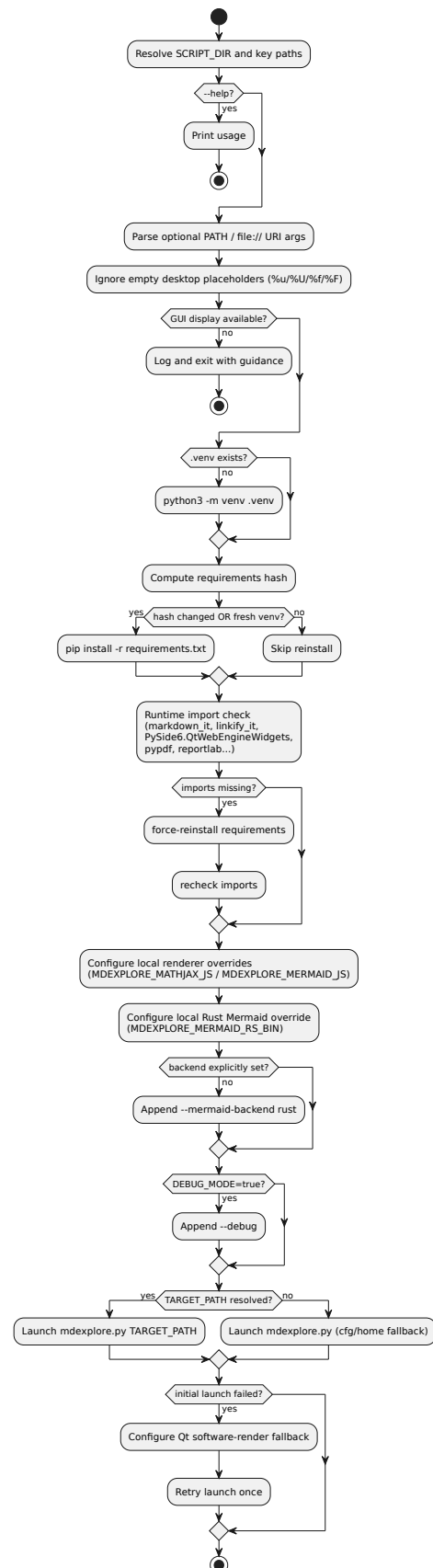
mdexplore UML

This document provides a subsystem-level PlantUML view of the current `mdexplore` implementation across `mdexplore.py`, `mdexplore.sh`, and `mdexplore_app/*`. All diagrams are embedded so they can be rendered directly by mdexplore (or any Markdown viewer with PlantUML support).

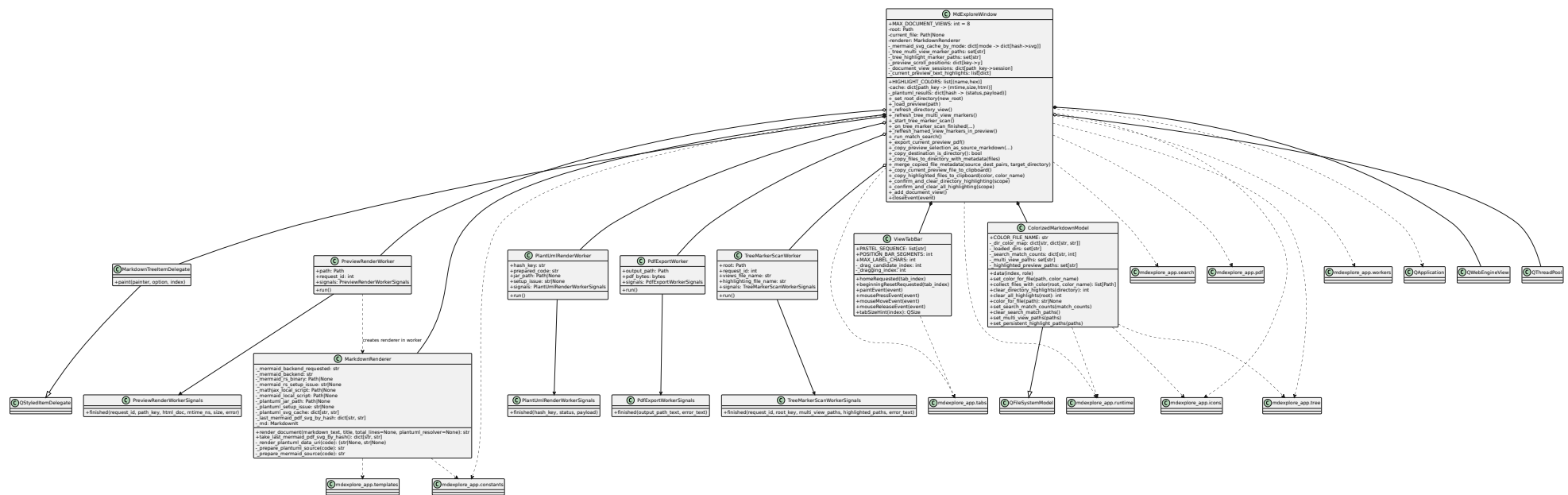
Detailed render/caching forks (GUI vs PDF, JS Mermaid vs Rust Mermaid, cache mode ownership, and restore behavior) are intentionally consolidated in the deeper render/debugging sections of `DEVELOPERS-AGENTS.md`. This UML file keeps those areas abstracted at system/class boundaries to avoid duplicating low-level render logic across two docs.

Low-risk modularization note: support code now lives in `mdexplore_app/` (`constants.py`, `runtime.py`, `search.py`, `templates.py`, `pdf.py`, `icons.py`, `file_coordination.py`, `workers.py`, `tree.py`, `tabs.py`), while the main orchestration, renderer/template wiring, and UI state machine remain in `mdexplore.py`. The diagrams below show that split at subsystem boundaries rather than expanding every extracted helper inline.

2. Launcher Runtime Activity (mdexplore.sh)

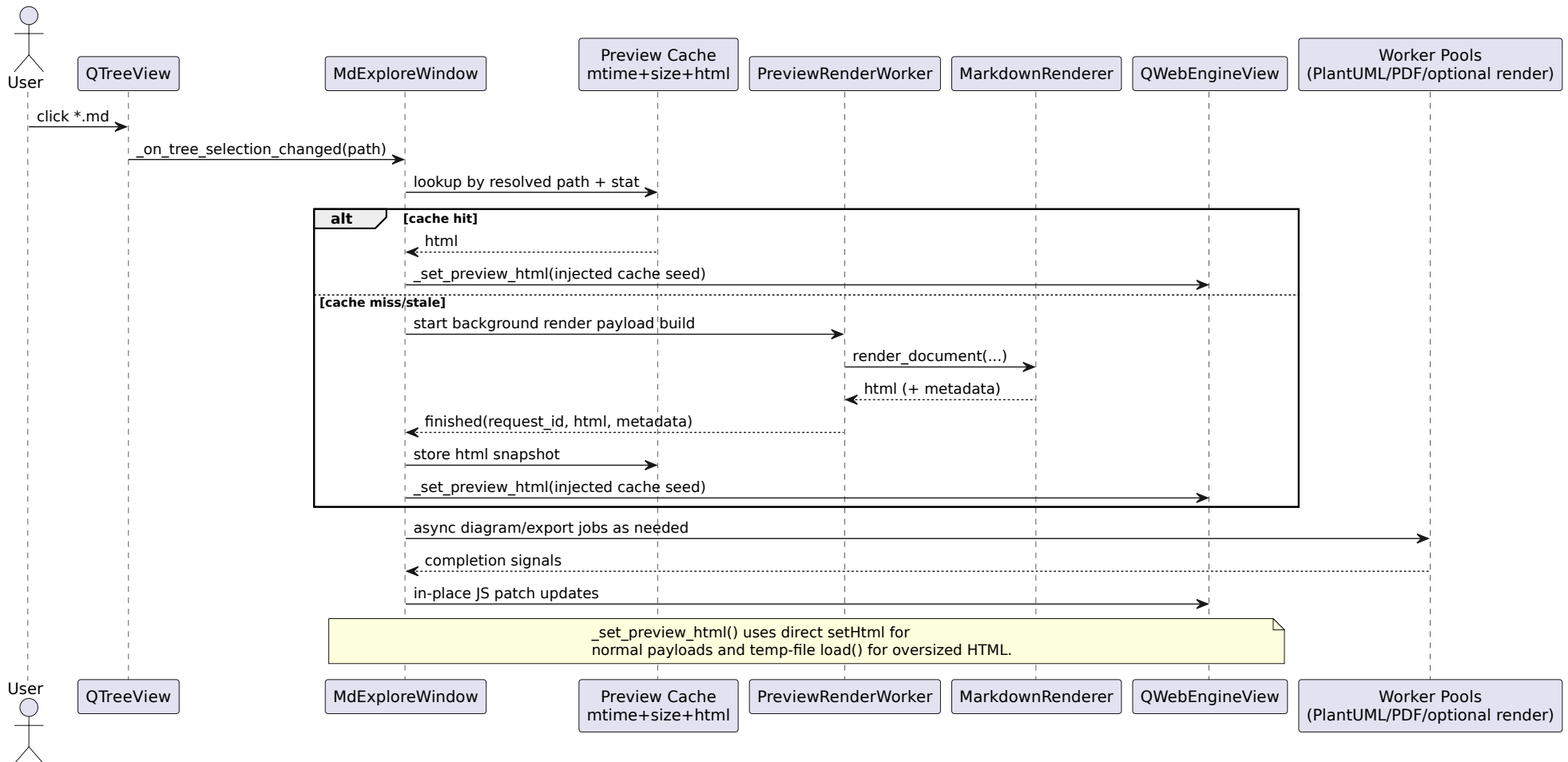


3. Core Class Diagram



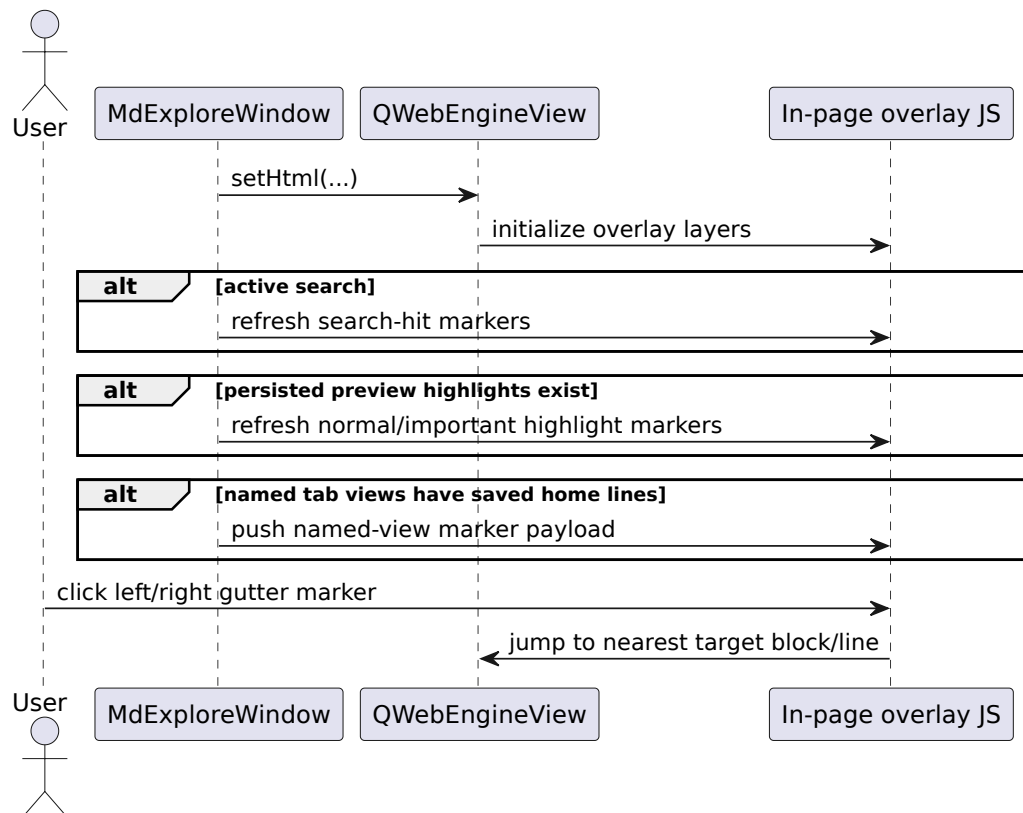
4. Preview Pipeline (Abstracted)

The detailed render/cache branches are covered in the render/debugging sections of `DEVELOPERS-AGENTS.md`. This section intentionally keeps an architectural boundary view only.

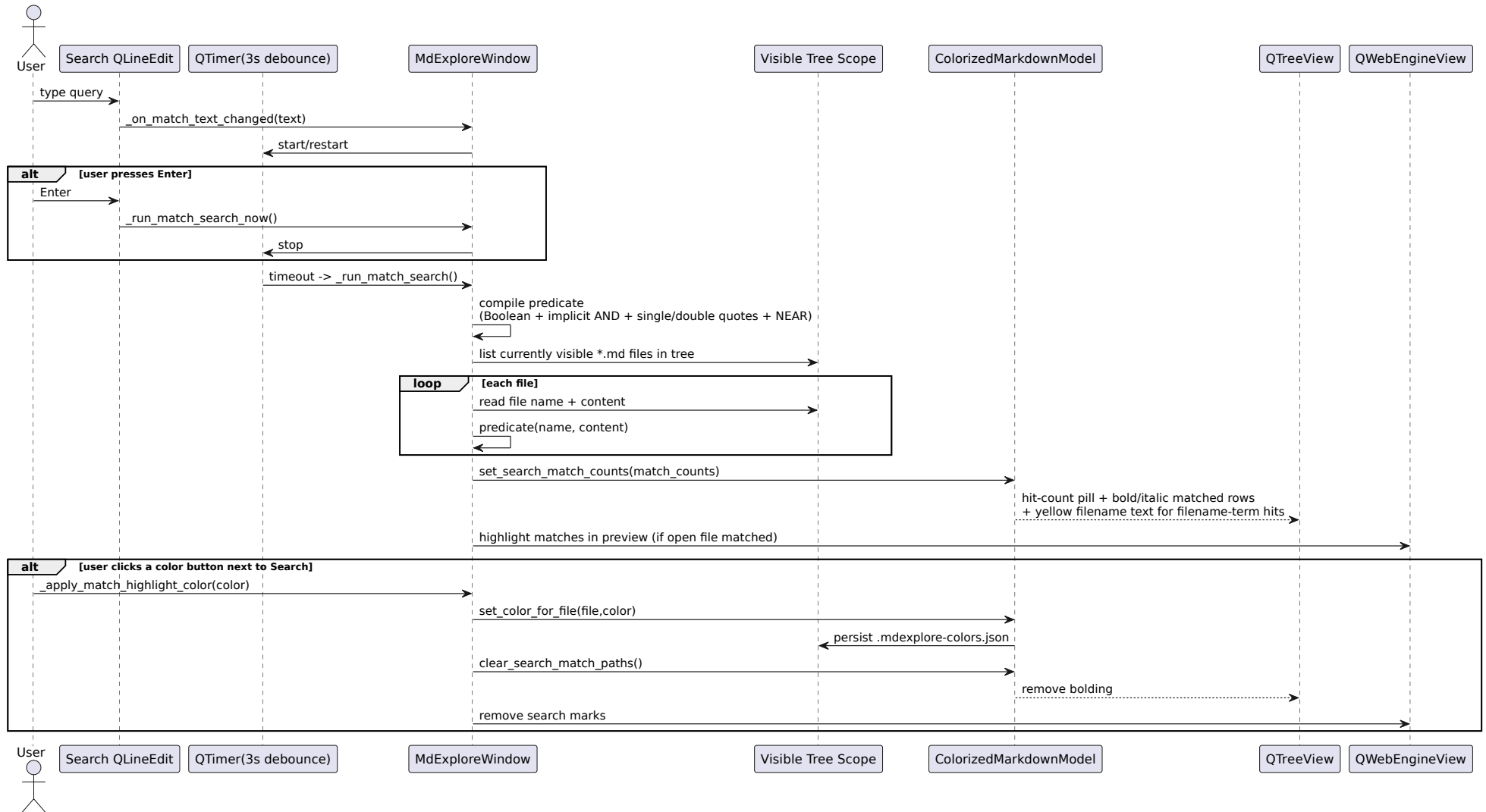


4.1 Preview Navigation Overlays

These are intentionally shown here only as boundary-level UI overlays. The exact DOM/CSS/JS mechanics remain in the deeper render/debugging sections of `DEVELOPERS-AGENTS.md`.



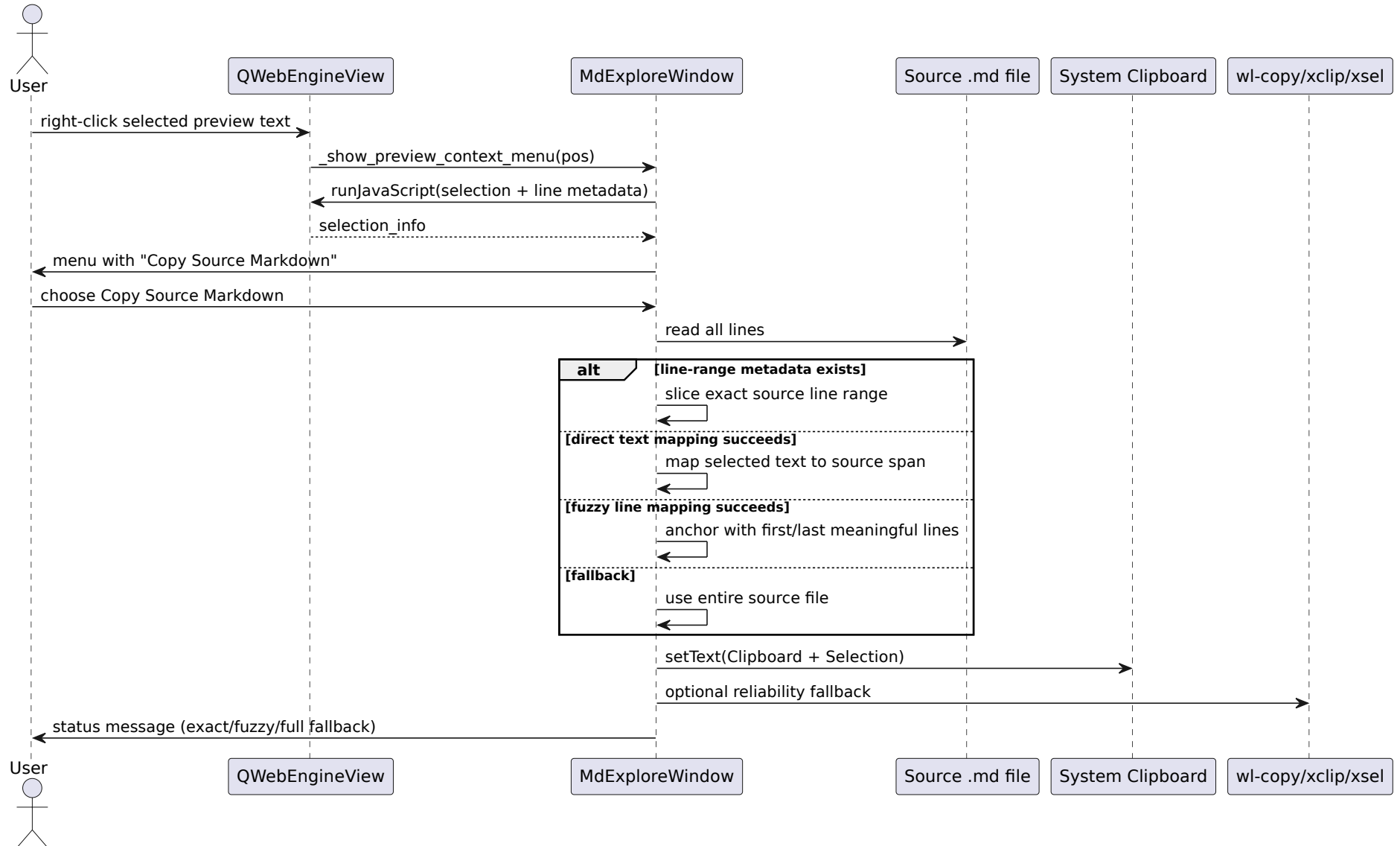
5. Search + Highlight + Apply Color Flow



Search semantics note:

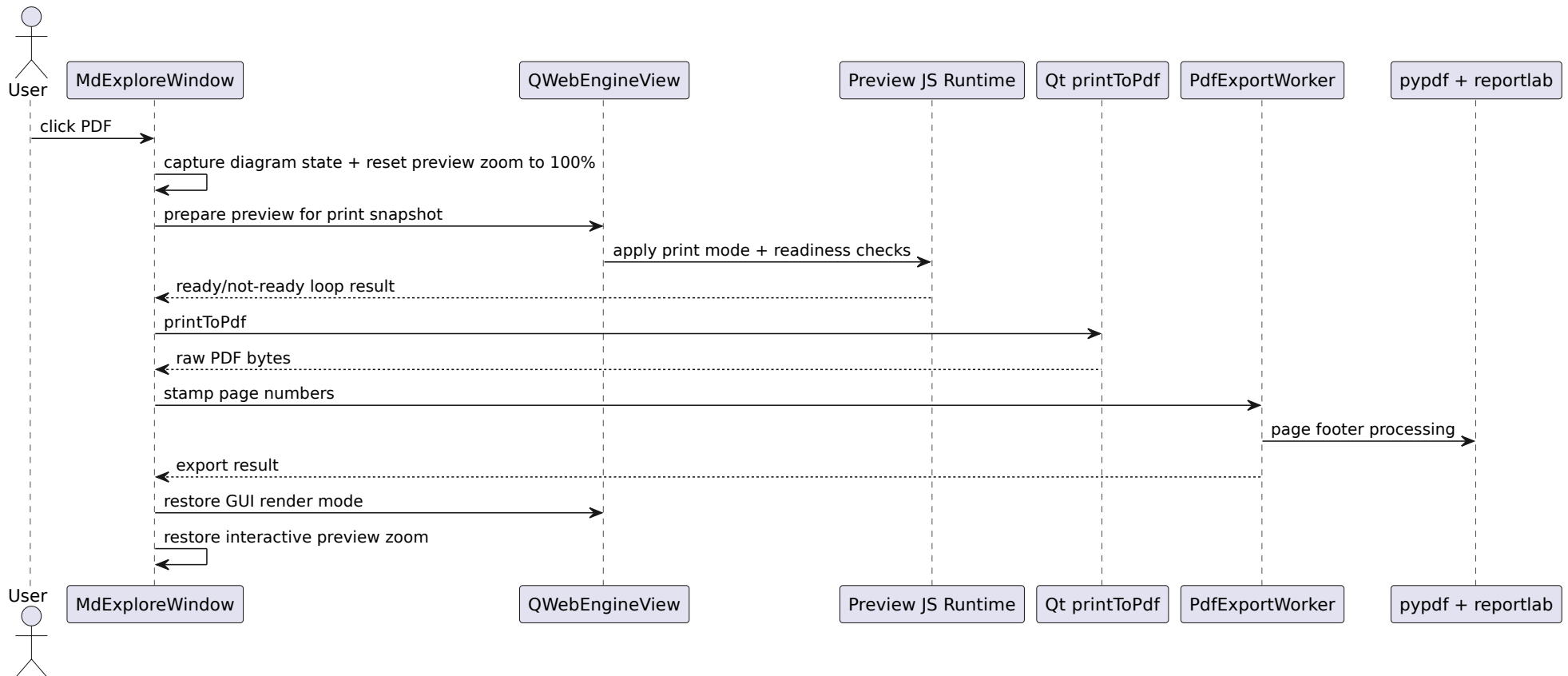
- unquoted terms are case-insensitive,
- double-quoted phrases are case-insensitive and preserve spaces,
- single-quoted phrases are case-sensitive and preserve spaces,
- only the opening quote character closes a quoted term, so apostrophes inside double-quoted phrases stay literal,
- `NEAR ( . . . )` requires distinct qualifying occurrences per term, and single-word NEAR terms use word boundaries for proximity matching.

6. Preview Context Menu: Copy Source Markdown

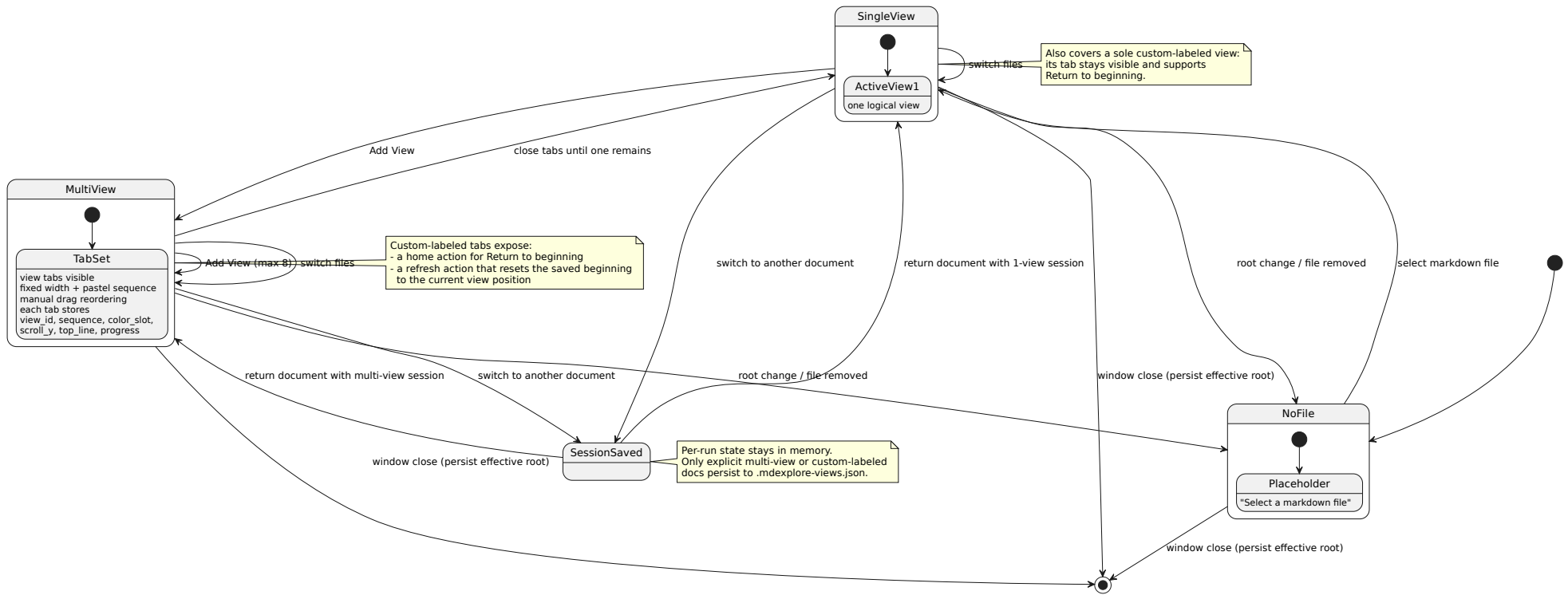


7. PDF Export Pipeline (Abstracted)

Detailed PDF mode branching (JS Mermaid grayscale path vs Rust Mermaid default-themed PDF cache path) is documented in the deeper render/debugging sections of `DEVELOPERS-AGENTS.md`.



8. Document View/Tab Lifecycle



Notes

- Diagrams are based on current code in `mdexplore.py`, `mdexplore.sh`, and `mdexplore_app/*`.
- Render/caching branch internals are intentionally abstracted here and documented in the deeper render/debugging sections of `DEVELOPERS-AGENTS.md` to keep a single authoritative deep map.
- Preview-only zoom (`Ctrl++`, `Ctrl+-`, `Ctrl+0`) is also intentionally kept out of the UML internals here because it is a `QWebEngineView` scale adjustment, not a separate renderer/cache branch.
- Preview gutter overlays (search-hit markers, persistent-highlight markers, named-view home markers) are similarly abstracted here and described in more detail in `DEVELOPERS-AGENTS.md`, because they are post-render navigation aids rather than renderer/cache forks.
- Search accepts canonical `NEAR(...)` syntax while continuing to treat `CLOSE(...)` as a backward-compatible alias normalized to `NEAR(...)` internally.
- Named-view gutter markers now route back through the same saved-view restore path as tab selection, so marker navigation and tab selection land on the same saved location.
- Effective-root directory styling is stateful in the tree:
 - bold aqua-blue when no active-search hits are under effective scope,
 - bold yellow with appended hit-count pill when active-search hits exist under effective scope.
- Top-right copy controls now include destination mode (`Clipboard` vs `Directory`); directory mode copies files into a chosen folder and merges copied-file metadata into destination `.mdexplore-*` sidecars.
- Headless regressions for saved-view restore, preview markers, and search quoting now live in `tests/test_preview_regressions.py` and `tests/test_search_query_syntax.py`, while template-asset regressions live in `tests/test_template_assets.py` and tab-bar layout regressions live in `tests/test_tab_bar_layout.py`.
- Worker/threadpool usage is intentionally separated by concern:
 - render pool (preview HTML jobs),
 - PlantUML pool (diagram jobs),
 - PDF pool (post-processing/stamping).
- PlantUML rendering is non-blocking in UI flow: placeholders are rendered first, then patched in place as jobs finish.
- TODO (known issue): diagram zoom/pan restore (Mermaid and PlantUML) is not yet consistently reliable when leaving a document and returning in the same app run.